

# Murty-Simon at n=29: Fan-free candidate proof

Reviewer edition 2 - hostile-audited dependency replacement

Paul Lenz research project; mathematical development and drafting with ChatGPT/Geeps

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## Abstract

This reviewer edition replaces the historical use of G. Fan's 1987 upper-density theorem by a direct order-specific exclusion of every larger edge count. Fan remains cited for attribution. The historical reviewer-v1 proof surface is preserved unchanged. The Fan-free assembly has passed an internal hostile coverage/integrity audit; independent mathematical and computational review remain open.

**Canonical claim.**  $e(G) \leq 210$ , equality exactly  $K(14, 15)$ .

**Logical source.** `project/reviews/n29/2026-09-09-fan-free-v2/PROOF.md`.

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## Murty-Simon at n=29: Fan-free candidate proof, edition 2

9 September 2026. Fan-free edition 2, built from the frozen 8 September candidate proof. Research directed by Paul Lenz; mathematical development, implementation and internal checking by ChatGPT/Geeps.

**Status:** candidate proof. Independent mathematical review, external computational reproduction and novelty assessment remain OPEN. Fan's 1987 theorem is cited historically but is not a logical dependency of this edition. Frozen historical source SHA-256: `b5e8f78a9175d3a00f3911e17a24f388d735555975eca30690b9cacfda494b27`. Fan-free upper-range component: `project/research/fan-free-fixed-orders/2026-09-09-v1/FAN_FREE_REDUCTION.md`.

### 1. Statement

Let  $G$  be a finite simple diameter-two edge-critical graph on 29 vertices. The candidate statement is

$$e(G) \leq 210 = \left\lfloor \frac{29^2}{4} \right\rfloor,$$

with equality if and only if

$$G \cong K_{14,15}.$$

Deleting an edge is understood to increase the diameter, with disconnected pairs assigned infinite distance.

The proof does **not** use the later general  $293/500$  candidate theorem. It uses the same published reductions and residual/quasi-edge lemmas already isolated in the  $n=27/n=28$  programme, one new fixed-order  $\Delta=16$  calculation, and a short witness argument for  $\Delta=15$ .

## 2. Fan-free upper reduction and the two edge counts

Historically, edition 1 used G. Fan's 1987 strict density estimate, as reported by Wang, to reduce the upper-bound problem to 211 edges. Fan remains cited for attribution, but **his theorem is not a logical dependency of edition 2.**

The direct replacement is `project/research/fan-free-fixed-orders/2026-09-09-v1/FAN_FREE_REDUCTION.md`. In summary, every edge count  $m \geq 212$  is excluded without Fan: degree sum handles  $\Delta \leq 14$ ; the witness-deficit inequality at  $\Delta = 15$  is monotone stronger above 211; fresh exact arithmetic closes  $\Delta = 16$  throughout  $m = 212 \dots 232$ ; the pointwise charging argument at  $\Delta = 17$  strengthens as  $m$  rises; the residual h-index argument likewise excludes  $\Delta = 18 \dots 27$ ; and  $\Delta = 28$  is the universal-vertex/star case. Thus any upper-bound counterexample has exactly 211 edges. Equality at 210 is treated separately, exactly as in edition 1.

The complete historical Fan-based proof remains preserved at `project/reviews/n29/2026-09-08-candidate-v1/PROOF.md`.

A bipartite graph of diameter two is complete bipartite, because a missing cross-part pair has odd distance at least three. Hence a bipartite graph on 29 vertices has at most  $14 \cdot 15 = 210$  edges, with equality only  $K(14, 15)$ .

Assume henceforth that  $G$  is non-bipartite. The published dominating-edge result of Dailly, Foucaud and Hansberg gives at most  $\lfloor (29^{2/4}) - 2 \rfloor = 208$  edges for a non-bipartite diameter-two-critical graph with a dominating edge; its exceptional graph has order six. Consequently neither dense edge count has a dominating edge.

A universal vertex forces an edge-critical graph to be a star: any edge among the remaining vertices could be removed while every pair stayed at distance at most two. The star case is therefore sparse and is handled separately below.

If  $\Delta \leq 14$ , the degree sum gives

$$2e(G) \leq 29 \cdot 14 = 406,$$

so  $e(G) \leq 203$ . Thus only  $\Delta \geq 15$  can occur at 210 or 211 edges.

## 3. Witness lemma and the complete $\Delta = 15$ argument

A **direct witness** is an edge  $uv$  whose endpoints have no common neighbour. A **two-step witness** is a nonedge  $uv$  whose endpoints have exactly one common neighbour.

Every critical edge is covered by one of these witnesses. Indeed, delete a critical edge  $xy$  and choose a pair  $u, v$  whose distance becomes greater than two. In  $G$  their distance is at most two and every path of length at most two must use  $xy$ . If  $u, v$  are adjacent, their edge is  $xy$  and they have no common neighbour, giving a direct witness. If  $u, v$  are nonadjacent, their unique length-two path uses  $xy$ ; a second common neighbour would give another length-two path after  $xy$  were deleted. Thus  $uv$  is a two-step witness. A direct witness covers its own edge; a two-step witness can cover at most the two edges of its unique length-two path.

Because there is no dominating edge, every witness pair  $uv$  satisfies

$$d(u) + d(v) \leq 28. \tag{3.1}$$

For a two-step witness,  $N(u)$  and  $N(v)$  lie among the other 27 vertices and have intersection one, so the degree sum is at most  $27 + 1 = 28$ . For a direct witness, the neighbourhoods are disjoint; if their union were all 29 vertices,  $uv$  would be a dominating edge.

Now suppose  $\Delta=15$  and put

$$\varepsilon_x = 15 - d(x), \quad L = \{x : \varepsilon_x \geq 2\}, \quad O = \{x : \varepsilon_x = 1\}.$$

Write  $h=|L|$  and  $o=|O|$ . By (3.1), every witness pair has deficit sum at least two. A witness entirely outside  $L$  must therefore have both endpoints in  $O$ . If

$$T = 29 \cdot 15 - 2e(G),$$

then

$$2h + o \leq T. \quad (3.2)$$

Let  $X=V(G)\setminus L$ . Count the edges having an endpoint in  $L$  directly, and cover the remaining  $X$ - $X$  edges by witnesses. A witness with both endpoints in  $L$  covers no  $X$ - $X$  edge. A missing cross pair between  $L$  and  $X$  covers at most one  $X$ - $X$  edge. A witness within  $O$  covers at most two. If  $e(L,X)$  is the number of actual cross edges, the number of missing cross pairs is  $h(29-h)-e(L,X)$ . Hence

$$\begin{aligned} e(G) &\leq \binom{h}{2} + e(L, X) + (h(29-h) - e(L, X)) + 2\binom{o}{2} \\ &= \binom{h}{2} + h(29-h) + o(o-1). \end{aligned} \quad (3.3)$$

No injective assignment of witness pairs to edges is assumed: each edge is assigned one witness, while the displayed capacities are upper bounds on how many edges any witness type can cover.

### 3.1 Excluding 211 edges at $\Delta=15$

At  $e(G)=211$ ,  $T=13$ . For each integer  $h=0, \dots, 6$ , (3.2) gives  $o \leq 13-2h$ . The right side of (3.3) is increasing in  $o \geq 1$ , so the largest possible bounds are

| $h$ | $o$ max | upper bound |
|-----|---------|-------------|
| 0   | 13      | 156         |
| 1   | 11      | 138         |
| 2   | 9       | 127         |
| 3   | 7       | 123         |
| 4   | 5       | 126         |
| 5   | 3       | 136         |
| 6   | 1       | 153         |

Every value is below 211. Thus  $\Delta=15$  is impossible at 211 edges.

### 3.2 Equality at 210 edges and $K(14,15)$

At  $e(G)=210$ ,  $T=15$ . The analogous table is

| $h$ | $o$ max | upper bound |
|-----|---------|-------------|
| 0   | 15      | 210         |

| h | o max | upper bound |
|---|-------|-------------|
| 1 | 13    | 184         |
| 2 | 11    | 165         |
| 3 | 9     | 153         |
| 4 | 7     | 148         |
| 5 | 5     | 150         |
| 6 | 3     | 159         |
| 7 | 1     | 175         |

Equality therefore forces  $h=0$  and  $o=15$ . All witnesses lie inside  $O$ . We can now distinguish actual  $O$ -edges from  $O$ -nonedges: a direct witness edge covers one edge, while a two-step witness nonedge covers at most two. Consequently

$$210 \leq e(G[O]) + 2 \left( \binom{15}{2} - e(G[O]) \right) = 210 - e(G[O]).$$

Thus  $O$  is independent. Its 15 vertices each have degree 14, and there are exactly 14 vertices outside  $O$ . Every vertex of  $O$  is therefore adjacent to every vertex outside  $O$ . Those  $15 \cdot 14 = 210$  cross edges exhaust  $E(G)$ . Hence

$$G = K_{15,14}.$$

Conversely  $K(14,15)$  is diameter-two edge-critical: after deleting a cross edge  $uv$ , its endpoints have no path of length two (the graph is bipartite) and do have a length-three path because both parts have size at least two. Thus the diameter increases to three.

This settles  $\Delta=15$  completely, without enumeration.

#### 4. Complement residual setup for $\Delta=16$

It remains to exclude  $\Delta=16$  at both 211 and 210 edges. Put  $H = \text{complement}(G)$ , choose a minimum-degree vertex  $v$  of  $H$ , and set

$$A = N_H(v), \quad B = V(H) \setminus N_H[v].$$

Then

$$|A| = a = 29 - 1 - 16 = 12, \quad |B| = b = 16.$$

Let  $C = H[A]$ ,  $F = \text{complement}(C)$  on  $A$ , and  $d_i = d_F(i)$ . For each missing unordered pair  $uw$  in  $H[B]$ , edge-criticality of  $G$  implies that  $H+uw$  contains a new adjacent total-dominating pair. Because  $u, w$  both miss  $v$ , that pair is not  $\{u, w\}$ ; after interchanging endpoints it is an existing cross edge  $ui$  whose open  $H$ -neighbourhoods cover every vertex except  $w$ . Write  $ui \rightarrow w$ . Choose one such cross edge for every missing  $B$ -pair. Selection is indexed by missing **unordered**  $B$ -pairs. For each such pair designate exactly one corresponding cross edge; opposite orientations of the same unordered pair cannot both be selected. Different selected edges therefore recover different unordered  $B$ -pairs from their  $B$ -source and unique exception. All other  $H[A, B]$  edges are residual.

Let  $\rho_u$  and  $R_i$  be residual row/column degrees,  $q_u$  and  $p_u$  outgoing/incoming selected-pair degrees, and  $x_i$  the actual selected degree of label  $i$ . Put

$$r = \sum_u \rho_u = \sum_i R_i, \quad s_i = \max(0, d_i - R_i), \quad S = \sum_i s_i, \quad t = e(G) - b(29 - b).$$

For the two scopes

$$(m, t) = (211, 3) \quad \text{and} \quad (210, 2).$$

Counting H gives

$$e(F) = r + t, \quad \sum_i d_i = 2(r + t), \quad S \geq r + 2t. \quad (4.1)$$

Because  $t > 0$ , the residual-activity lemma applies: every B-source has  $\rho_u = 1$ . The selected-edge injection gives

$$d_i \leq \rho_u + R_i, \quad d_i \leq \rho_u + \rho_w, \quad \rho_w + q_w \geq q_u - 1, \quad (4.2)$$

whenever  $u \rightarrow w$  is selected. The actual selected neighbourhood also gives

$$R_i + x_i \geq q_u + p_u. \quad (4.3)$$

The source and supplement capacities are

$$q_u + \rho_u \leq a, \quad p_u \leq \rho_u + (b - a - 1) = \rho_u + 3, \quad q_u + p_u \leq b - 1. \quad (4.4)$$

These are the same graph lemmas used by the order-28 direct197 route, with  $b=16$  and the displayed  $t$  supplied explicitly. The adapted programs infer  $a$  and  $b$  from the current demand/row lengths and do not retain a hard-coded  $b=15$  assumption in the joint/LP stages.

The inherited charging inequality yields the necessary demand condition

$$\sum_{i=1}^{12} \frac{s_i(13 - 2s_i)}{12 - s_i} \geq 16 + 2t. \quad (4.5)$$

The finite calculation deliberately over-enumerates arithmetic systems satisfying these necessary conditions; a numerical survivor is never treated as a graph.

## 5. Complete Delta=16 finite calculation

The adaptation starts from the hash-pinned  $n=28$  direct197 source archive. Only the fixed scope changes to  $a=12, b=16$  and  $t$  in  $\{3, 2\}$ . The demand generator independently checks its complete integer domain. Two separately compiled C++ residual-row implementations produce byte-identical survivor and band files. The projected pair screen is regenerated from the current  $t$ . The joint-column propagation is run twice: the discovery implementation and a separately compiled/set-valued checker agree on every surviving state. Final infeasibility is accepted only when an integer Farkas certificate is verified against a separately reconstructed named linear system.

A floating-point solver is used only to *propose* certificates. Solver status alone is never an exclusion. Every accepted certificate has nonnegative integer multipliers on inequalities and signed integer multipliers on equalities, and the checker verifies coefficientwise nonnegativity together with a strictly negative right-hand side.

### 5.1 The 211-edge scope: $t=3$

| Stage                                    | Input / retained |
|------------------------------------------|------------------|
| Complete demand tuples                   | 4,867            |
| Demand tuples retained                   | 339              |
| Residual rows enumerated                 | 1,848,957        |
| Residual row survivors                   | 206              |
| Projected survivors                      | 118              |
| Joint-column survivors                   | 36               |
| Shared-adjacency exact contradictions    | 13               |
| Source-degree-typed exact contradictions | 23               |
| Endpoint-type contradictions needed      | 0                |
| <b>Final survivors</b>                   | <b>0</b>         |

The residual row implementations agree on all 1,848,957 rows. The projected stage excludes 47 zero-slack rows and 41 pair-threshold rows. The joint stage excludes 38 by source matching, two by source Hall, and 42 by joint total-source capacity, leaving 36. All 36 then have exact Farkas contradictions.

### 5.2 The 210-edge scope: $t=2$

| Stage                                     | Input / retained |
|-------------------------------------------|------------------|
| Complete demand tuples                    | 9,251            |
| Demand tuples retained                    | 901              |
| Residual rows enumerated                  | 5,765,218        |
| Residual row survivors                    | 2,087            |
| Projected survivors                       | 1,225            |
| Joint-column survivors                    | 593              |
| Shared-adjacency exact contradictions     | 213              |
| Source-degree-typed exact contradictions  | 378              |
| Actual-endpoint-type exact contradictions | 2                |
| <b>Final survivors</b>                    | <b>0</b>         |

The residual implementations agree on all 5,765,218 rows. The projected stage excludes 734 zero-slack rows and 128 pair-threshold rows. The independent joint checks agree on all 1,225 inputs and leave 593. Their exclusions are 311 source-matching, 29 source-Hall, six empty label domains, 58 total-source, 222 joint-total-source, and six pair-Hall contradictions. The exact LP stages reject all 593 remaining rows, including the last two only after the actual endpoint-degree model.

Thus Delta=16 is impossible at both 211 and 210 edges.

## 6. Delta=17 by the pointwise charging bound

For Delta=17 we have  $a=11$ . The same charging inequality requires

$$17 + 2t \leq \sum_{i=1}^{11} \frac{s_i(12 - 2s_i)}{11 - s_i}.$$

For every integer  $0 \leq s \leq 10$ ,

$$\frac{s(12-2s)}{11-s} \leq \frac{16}{7}.$$

Indeed

$$\frac{16}{7} - \frac{s(12-2s)}{11-s} = \frac{2(s-4)(7s-22)}{7(11-s)} \geq 0$$

at every integer  $s=0, \dots, 10$ . Hence the right side is at most

$$11 \cdot \frac{16}{7} = \frac{176}{7} < 29.$$

At 210 edges,  $t=210-17 \cdot 12=6$  and the required left side is 29. At 211 edges,  $t=7$  and it is 31. Both are impossible.

### 7. Delta from 18 through 27 by residual h-index

Let  $h$  be the largest integer for which at least  $h$  residual rows have degree at least  $h$ . The source-demand injection implies  $s_i \leq h$  for every label: a label demanding  $s_i$  selected sources requires  $s_i$  distinct sources with residual degree at least  $s_i$ .

Since  $t > 0$ , every B-row is residual-active. Therefore

$$r \geq h^2 + (b - h) = b + h(h - 1).$$

On the other hand  $S \leq ah$  and  $S \geq r + 2t$ . Consequently

$$b + 2t \leq (a + 1)h - h^2 \leq \left\lfloor \frac{(a + 1)^2}{4} \right\rfloor = \left\lfloor \frac{(29 - b)^2}{4} \right\rfloor. \quad (7.1)$$

For  $m=210$  and  $b=18, \dots, 27$  the left/right pairs begin with  $42 > 30$  and then become still more separated. For  $m=211$  they begin with  $44 > 30$ . The complete exact table is reproduced by the supplied arithmetic checker. Thus no  $b$  in  $18, \dots, 27$  is possible at either dense edge count.

For  $\Delta=28$ , the universal-vertex argument gives a star, which has only 28 edges.

### 8. Assembly

The Fan-free upper-range reduction excludes every  $m \geq 212$ ; it remains only to exclude  $m=211$  and analyse equality at  $m=210$ .

At  $m=211$ :

- $\Delta \leq 14$  is impossible by degree sum;
- $\Delta=15$  is excluded by the witness count in Section 3;
- $\Delta=16$  is excluded by the complete direct calculation in Section 5;
- $\Delta=17$  is excluded by Section 6;
- $18 \leq \Delta \leq 27$  is excluded by Section 7;
- $\Delta=28$  gives a star.

Therefore  $e(G) \leq 210$ .

Suppose  $e(G)=210$ . Bipartite graphs already give equality only at  $K(14,15)$ . In the non-bipartite case the same degree partition applies.  $\Delta \leq 14$ ,  $\Delta=16$ ,  $\Delta=17$ ,  $\Delta \geq 18$  and the star case are all excluded. Thus  $\Delta=15$ , and Section 3.2 forces  $G=K(14,15)$ , contradicting the assumption that it was non-bipartite. Hence globally equality occurs exactly for  $K(14,15)$ .

This proves the stated **candidate** order-29 result, subject to the explicit external and universal-lemma review obligations below.

## 9. Audit boundaries and provenance

The  $\Delta=16$  source is an adaptation of `project/research/general_n/2026-09-07-direct-197-v8/MurtySimon_N28_197_Direct_v8.zip`, SHA-256 `b753076ffc755066a0c57756be91cc5b265c163d869244d3aa65e3943799347b`. The first hosted  $n=29$  replay attempts exposed environment/setup issues (missing SciPy, then isolated Python hiding a user-site install, then missing Boost headers). Those failures are preserved as failed executions and are not mathematical counterexamples. The fourth hosted run, GitHub Actions run `34220977858` at commit `42954d3094860a218e6ba8a418c35b7061449b31`, completed successfully on Ubuntu 24.04 after pinning SciPy 1.17.0 in an isolated virtual environment and installing Boost headers. Its aggregate output is reproduced verbatim in the review package. The three earlier failed executions are retained as environment/setup failures rather than rewritten as mathematical runs.

The fixed-order arithmetic is exhaustive over its stated numerical relaxation, not over all labelled graphs. Actual-graph regressions elsewhere in the project contain no positive-surplus critical graph, so the universal graph-to-model implication rests on the written lemmas rather than empirical sampling. The solver-assisted discovery and exact checking implementations were written by the same assistant; clean-runner execution is not independent authorship or peer review.

Dailly–Foucaud–Hansberg’s published dominating-edge theorem remains an external input and is not re-proved here. Fan’s theorem is retained only as historical attribution and is not used logically in edition 2. The quasi-edge, residual-injection, activity, charging and h-index arguments are internal hand proofs with separate earlier audits and limited local formalisation, not a full Lean proof of this order-29 theorem.

No order above 29, unrestricted all-order Murty–Simon solution, novelty determination, priority claim or external endorsement is asserted by this manuscript.

## References

- [1] Tao Wang, *On Murty-Simon Conjecture*, arXiv:1205.4397 (2012), especially page 2 for the strict Fan bound as reported there. <https://arxiv.org/pdf/1205.4397>
- [2] Antoine Dailly, Florent Foucaud and Adriana Hansberg, *Strengthening the Murty-Simon conjecture on diameter 2 critical graphs*, arXiv:1812.08420 (2018), Theorem 4 for the non-bipartite dominating-edge bound. <https://arxiv.org/pdf/1812.08420>
- [3] Project direct197-v8 proof and exact source package: `project/research/general_n/2026-09-07-direct-197-v8/`. The  $n=29$   $\Delta=16$  calculation uses its hash-pinned source archive as an implementation antecedent but generates fresh  $n=29$  domains.
- [4] Project  $n=27$  candidate proof: `project/reviews/n27/2026-09-07-candidate-v1/PROOF.md`, for the earlier full written derivations of the residual-activity and h-index framework. The relevant arguments are restated above at the level needed for  $n=29$ .